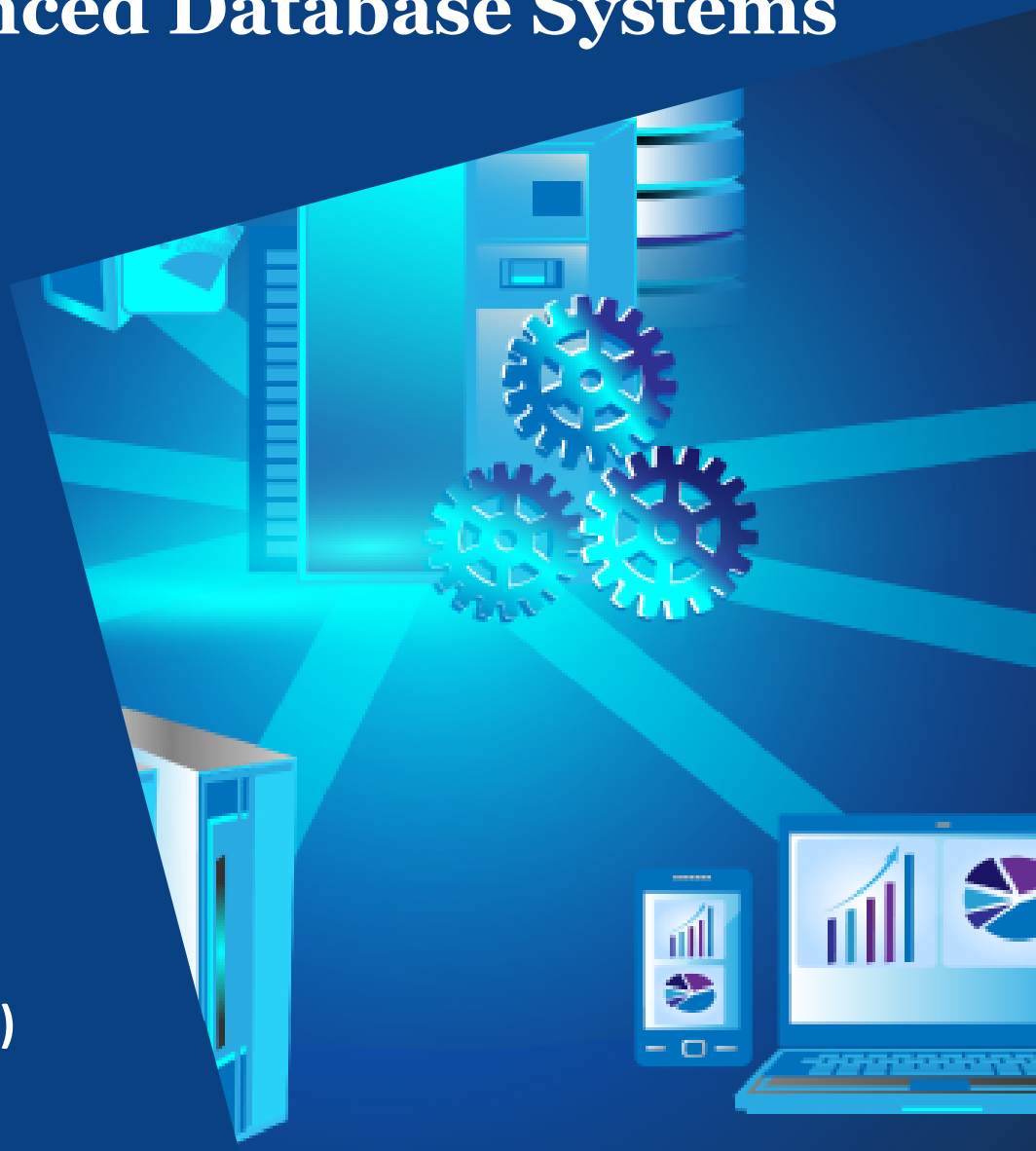
THE UNIVERSITY OF MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026



Lecturer: Farhana Choudhury (PhD)

Live lecture – Week 4

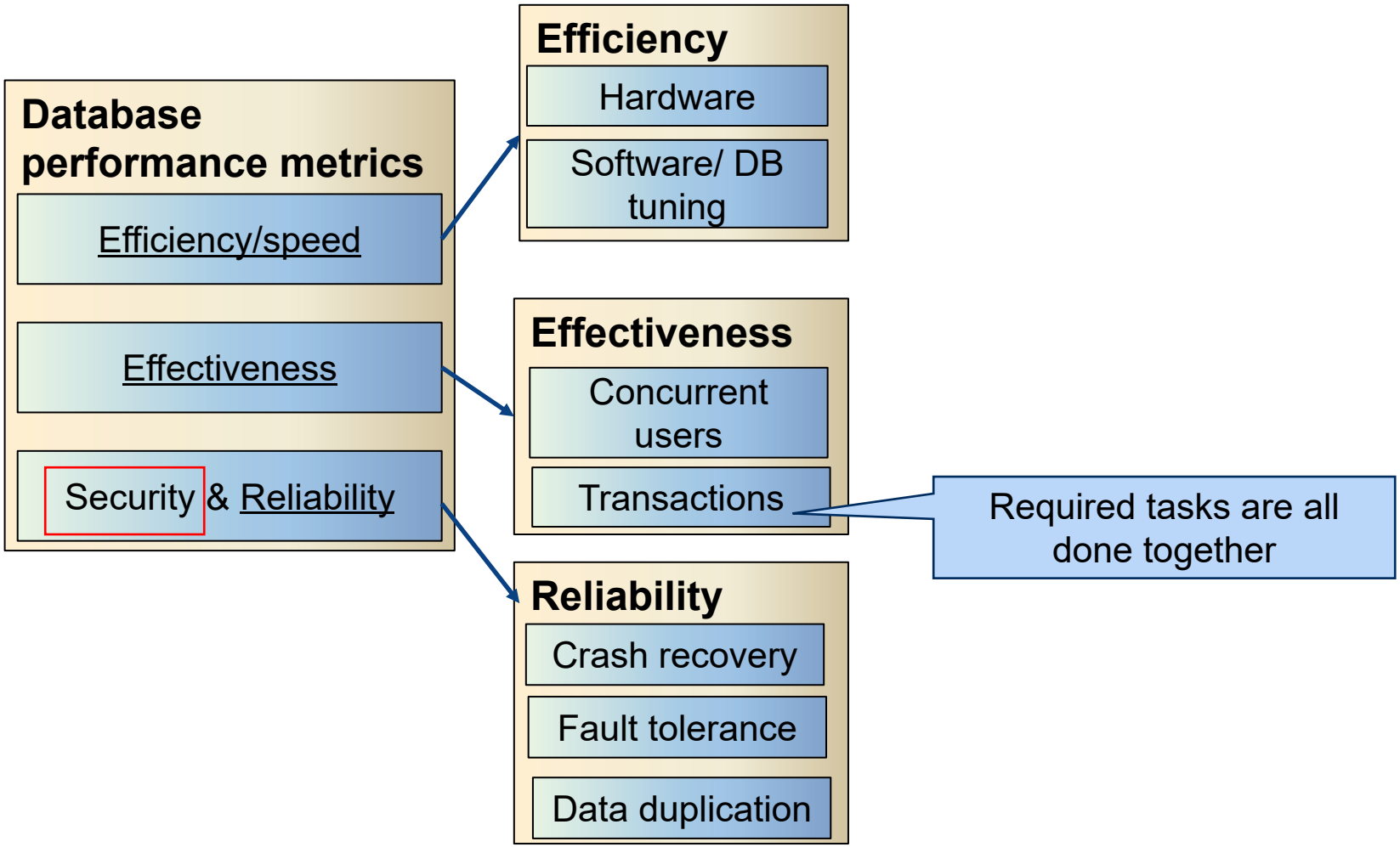
Quizzes

Quiz 5 next week!

Contents of Week 2 and Week 3 lectures and corresponding tutorials.

[Not Week 5 tutorial contents]

Today's lecture



This week's contents

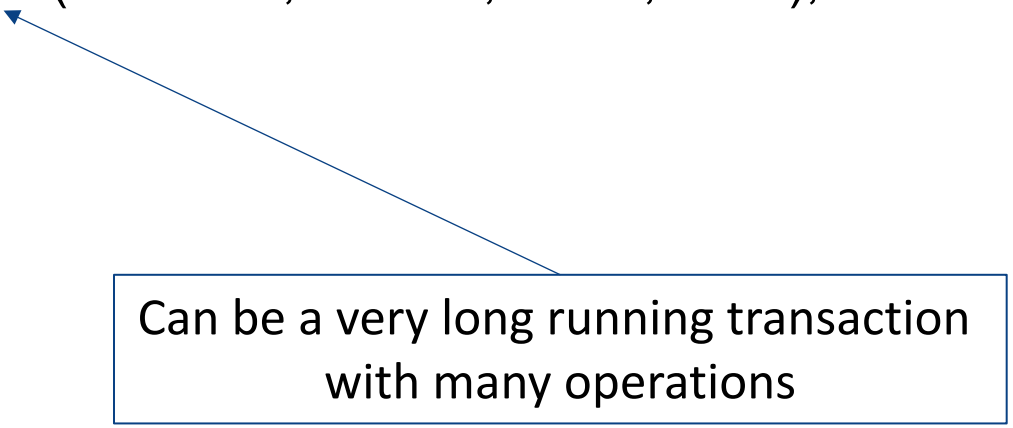
Transactions **the building block of databases as consistency, concurrency, crash recovery – all these powerful and important database properties are developed based on transaction concepts**

- Some related concepts (connecting SQL with a programming language, variable types, etc.)
- Types of transactions
  - Flat transactions
  - Flat transactions with save points
  - Nested transactions
- Transaction processing monitor

Flat Transaction

Everything inside BEGIN WORK and COMMIT WORK is at the same level; that is, the transaction will either survive together with everything else (commit), or it will be rolled back with everything else (abort)

```
exec sql BEGIN WORK;
    AccBalance = DodebitCredit(BranchId, TellerId, AccId, delta);
    send output msg;
exec sql COMMIT WORK;
```



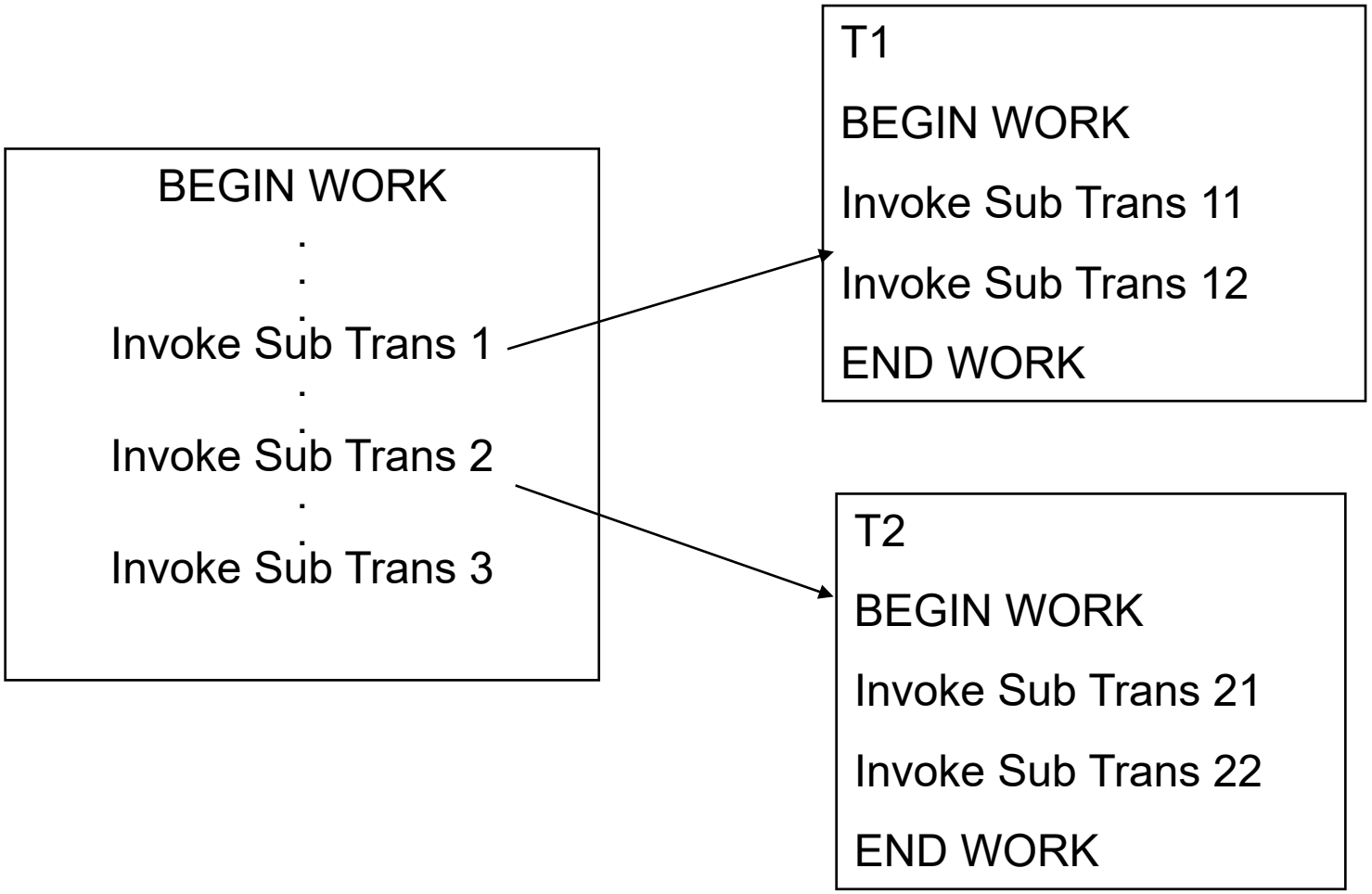
```
/* Withdraw money -- bank debits;  Deposit money – bank credits */
Long DoDebitCredit(long BranchId,
            long TellerId, long AccId, long AccBalance, long delta){
   exec sql UPDATE accounts
   SET AccBalance =AccBalance + :delta
   WHERE AccId = :AccId;
   exec sql SELECT AccBalance INTO  :AccBalance
   FROM accounts WHERE AccId = :AccId;
   exec sql UPDATE tellers
   SET TellerBalance = TellerBalance + :delta
   WHERE TellerId = :TellerId;
   exec sql UPDATE branches
   SET BranchBalance = BranchBalance + :delta
   WHERE BranchId = :BranchId;

   Exec sql INSERT INTO history(TellerId, BranchId, AccId, delta, time)
   VALUES( :TellerId, :BranchId, :AccId, :delta, CURRENT);
return(AccBalance);
}
```

Limitations of Flat Transactions?

Solution – flat transaction with savepoints

Resources:
https://dev.mysql.com/doc/refman/9.0/en/savepoint.html

Nested Transactions



Nested Transactions

Commit rule

- A subtransaction can either commit or abort, however, **commit cannot take place unless the parent itself commits.**

- Subtransactions have A, C, and I properties but not D property unless all its ancestors commit.

- Commit of a sub transaction makes its results available only to its parents.

Roll back Rules

If a subtransaction rolls back, all its children are forced to roll back.

What are the advantages?

Time for a poll - Pollev.com/farhanachoud585

Embedded SQL example in C

(Open Database Connectivity)

```
int main()
{       exec sql INCLUDE SQLCA; /*SQL Communication Area*/
        exec sql BEGIN DECLARE SECTION;
          /* The following variables are used for communicating
                between SQL and C */
          int OrderID; /* Employee ID (from user) */
          int CustID; /* Retrieved customer ID */
          char SalesPerson[10] /* Retrieved salesperson name */
          char Status[6] /* Retrieved order status */
        exec sql  END DECLARE SECTION;
/* Set up error processing */
        exec sql WHENEVER SQLERROR GOTO query_error;
        exec sql WHENEVER NOT FOUND GOTO bad_number;
```

Proper error handling is important!

What can go wrong in this example?

(Open Database Connectivity)

```
int main()
{       …..//code to get data from database
        Printf("%d", ((get_salary(EMPID)/80000);

/* Set up error processing */
        WHENEVER DIVIDE_BY_ZERO GOTO inf_error;
        WHENEVER NOT_PERMITTED GOTO permission_err;
        WHENEVER NOT_FOUND GOTO bad_number;
```

Time for a poll - Pollev.com/farhanachoud585

Proper error handling is important! - Inferential attack/data leak

Transaction Processing Monitor

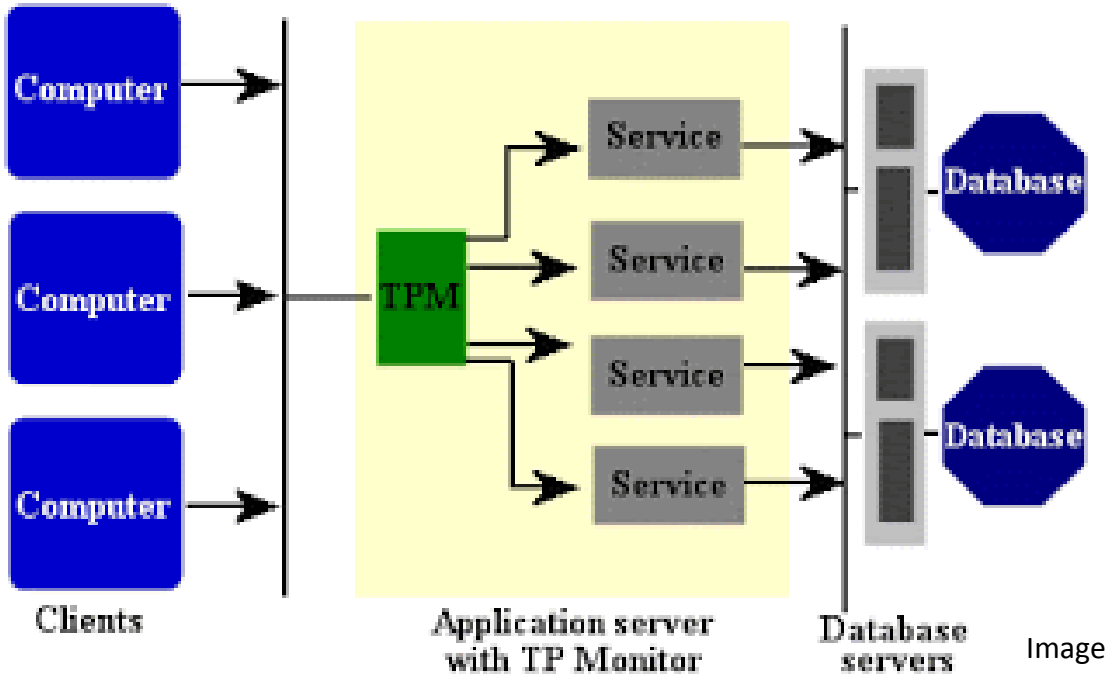


Image source: http://3.bp.blogspot.com

I*ntegrates* other system components and manage resources.

- Manages the transfer of data between clients and servers
- breaks down applications or code into transactions and ensures that all databases are updated properly
- It also takes appropriate actions if any error occurs

Core Concepts of Database management system